

FairCert: Certifiable Error Rate Fairness for ML and LLM Decision Systems

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A small FPR gap on one test set does not certify a classifier as fair. A point estimate has no confidence bound, so the observed gap may just be sampling noise. The issue is worst on intersectional subgroups, where the number of samples in per-group is small. Existing statistical fairness audits are not designed for LLM decision systems, nor do they pair certification with LLM-guided intersectional subgroup discovery. We present *FairCert*, a black-box assessment framework that produces finite-sample certificates on false positive rate (FPR) and true positive rate (TPR) gaps. It supports binary and multi-class settings. The main certificate uses Clopper-Pearson exact intervals with a Bonferroni correction across K intersectional groups. Besides certification, we add two supplementary tools. A permutation test shuffles group labels among the true negatives and recomputes the gap. Restricting to negatives keeps the Type I error rate valid when base rates differ across groups, and a bootstrap diagnostic serves as an exploratory alternative. We test FairCert on eight datasets covering finance, criminal justice, and vision. The evaluation covers ML classifiers, seven open-source LLMs from 3B to 30B parameters, and image models. Our experiments show that LLM fairness is domain dependent. Mistral-24B has an FPR gap of 0.259 on COMPAS but only 0.009 on ACS Income. Evaluating one attribute at a time hides intersectional gaps. On Credit Default dataset, certification passes for a single sensitive attribute but fails in the intersectional settings.

Keywords: Algorithmic fairness, Certification, False positive rate, LLM auditing, EU AI Act

1 Introduction

AI systems deployed in finance, criminal justice, and healthcare are increasingly used to make decisions about people, and they frequently produce decisions whose error rates vary across demographic groups [2, 23]. A credit scoring model that falsely denies loans, or a recidivism predictor whose false positive rate on Black defendants exceeds that on other groups, raises concerns under the EU AI Act [25, 26]. The Act requires the assessment but does not specify any particular fairness metric or evaluation methodology, and the GDPR adds further safeguards on automated decisions (Art. 22) and on data accuracy that a practical bias assessment would help document [20], yet bridging non-discrimination law to a technical fairness criterion remains contested [28].

Researchers evaluating these systems can report point estimates of the FPR and TPR gap with no statistical bound, leaving open whether an observed gap reflects a real population difference or sampling noise on a finite evaluation set. A growing line of work does attach statistical uncertainty to fairness metrics, through permutation tests [9], bootstrap-based simultaneous inference [6], and confidence intervals for disparate impact [3]. However,

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existing certification methods either demand white-box access to the model [21] or restrict to individual fairness and single-attribute settings [4].

At the same time, large language models are now being used as classifiers in tabular settings [16, 29], but prior studies have not produced finite-sample FPR or TPR parity certificates for these models, even though FPR and TPR matter when a false accusation or a missed detection causes harm.

In this work, we introduce **FairCert**, a black-box assessment framework that produces finite-sample certificates on FPR and TPR gaps across demographic groups for tabular ML classifiers, LLMs, and image models in both binary and multi-class settings. FairCert combines five modules into one pipeline, a stratified data split with a held-out audit set, LLM-guided intersectional subgroup discovery, per-group black-box error-rate evaluation on the held-out audit set, finite-sample certification using Clopper-Pearson exact intervals with Bonferroni correction, and threshold calibration with structured EU AI Act compliance reporting.

Contributions: Our key contributions are:

- **A certifiable fairness assessment framework:** We propose FairCert, which provides finite-sample certificates on FPR and TPR gaps using Clopper-Pearson intervals with Bonferroni correction. The framework enforces anti-leakage guarantees through a held-out audit set and supports LLM-guided subgroup discovery that shows disparities invisible to standard analysis (Section 4.1).
- **Additional exploratory methods:** Building on classical permutation inference [12, 19], we supplement the Bonferroni certificate with two exploratory alternatives. The permutation test permutes group labels within the $Y = 0$ subpopulation (not the full sample), which is necessary to avoid inflated Type I error when base rates differ across groups. The bootstrap estimates sampling variability without assuming the null, it serves as an exploratory diagnostic rather than a formal certificate (Section 3).
- **Multi-class extension:** We extend the framework beyond binary classification, each certifiable with the existing Clopper-Pearson machinery. We demonstrate this on 3-class income prediction (Section 4.1.5).
- **Evaluation across finance and justice domains:** We conduct experiments, evaluating ML classifiers and seven LLMs (3B to 30B parameters) across eight datasets including credit scoring, recidivism, and income prediction. On Credit Default, intersectional certification ($K = 4$) reveals disparities that single-attribute analysis misses. Moreover, LLM fairness gaps are task-dependent, near-certifiable on credit scoring but severe on criminal justice tasks.

The code and supplementary material can be found in <https://github.com/fahim-sikder/Certificable-Error-Rate-Fairness>.

2 Related Work

Fairness certification: Hardt et al. [14] introduced equalized odds, and Chouldechova [7] proved that equal PPV (predictive parity), equal FPR, and equal FNR cannot all be satisfied simultaneously when base rates differ across groups. Ruoss et al. (LCIFR) [21] verify individual fairness but require white-box access to the model. LLMCert-B [4] provides black-box probabilistic certificates for counterfactual LLM bias at the prompt level, not for tabular FPR/TPR gaps. Conformal prediction has also been adapted for equalized-coverage guarantees [30], but not for error-rate parity certification. Recent work quantifies statistical uncertainty in fairness metrics rather than

Table 1. Comparison of FairCert with existing fairness assessment approaches.

Method	Black-box	Certificates	Intersectional	LLM audit	Image
LCIFR [21]	✗	✓	✗	✗	✗
LLMCert-B [4]	✓	✓	✗	✓	✗
Xian et al. [29]	✓	✗	✗	✓	✗
HInter [27]	✓	✗	✓	✓	✗
FairCert (ours)	✓	✓	✓	✓	✓

certifying a tolerance. DiCiccio et al. [9] test fairness by permutation, Cherian and Candès [6] give bootstrap-based simultaneous confidence guarantee.

LLMs on tabular data: TabLLM [16] demonstrated that LLMs can perform tabular classification effectively. Cherepanova et al. [5] studied group fairness of LLMs on tabular data through in-context learning, and Xian et al. [29] evaluated black-box LLMs under group fairness criteria, but neither provides finite-sample statistical guarantees on the observed gaps.

Subgroup fairness and discovery: Kearns et al. [18] introduced the “fairness gerrymandering” problem and proposed testing over rich subgroup families, while Hebert-Johnson et al. [15] formalized multicalibration across intersectional groups. Prior work has also promoted intersectional fairness through knowledge distillation [24]. HInter [27] generates intersectional test inputs via mutation analysis with dependency parsing and metamorphic oracles. In contrast, FairCert does not introduce new statistical tools. It assembles known ones (Clopper-Pearson, Bonferonni, permutation tests) into a black box, LLM-integrated audit pipeline.

Regulation: The EU AI Act classifies credit scoring (Annex III, Area 5b) and administration of justice (Area 8a) AI as high-risk, requiring conformity assessments (Article 43) [25], but mandates no specific fairness metrics [26]. Wachter et al. [28] argue that EU non-discrimination law does not directly translate to any single statistical fairness criterion, and Hacker & Passoth [13] trace how transparency and explanation obligations under the GDPR and AI Act interact. Hildebrandt [17] frames the broader project of “legal protection by design.” FairCert targets this by producing verifiable statistical evidence of error-rate parity with sample-size and confidence semantics. Table 1 compares FairCert with existing methods.

3 FairCert

In this section, we present FairCert, a five-stage pipeline that evaluates classifiers for error-rate fairness and produces formal certificates. Figure 1 shows an overview of FairCert. FairCert supports both binary and multiclass classification settings (more details in Section 3.6).

Fairness definitions: It is challenging to give a single definition of fairness as this can be context-dependent. In this context, we focus on False-positive rate parity and equalized odds [14]. A classifier h satisfies FPR parity if $\Pr(h(X) = 1 \mid Y = 0, S = s) = \Pr(h(X) = 1 \mid Y = 0, S = s')$ for all groups s, s' . Equalized odds additionally requires equal TPR. For m protected attributes, intersectional groups $g \in S_1 \times \cdots \times S_m$ grow combinatorially, requiring stricter multiple comparison corrections.

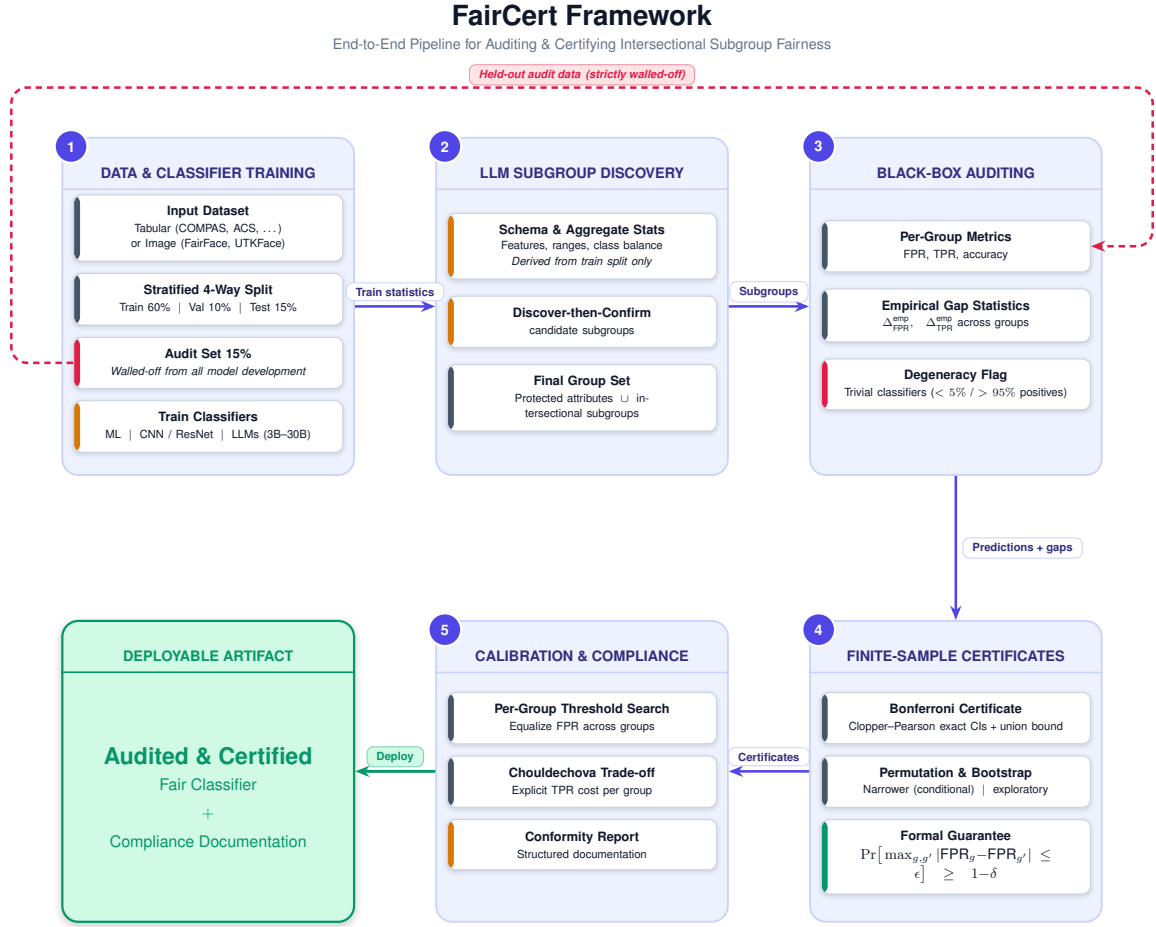


Fig. 1. The proposed FairCert architecture. Data is split into four stratified partitions with a held-out audit set (red). Classifiers are trained on the train split only. An LLM discovers intersectional subgroups (gold) from aggregate training statistics. The audit set flows exclusively to the black-box auditor, which computes per-group FPR/TPR. Certificates (green) formally certify whether gaps are within tolerance ϵ . Failed classifiers undergo threshold calibration with specific TPR cost reporting.

3.1 Framework Overview

FairCert consists of five stages: (1) data splitting with anti-leakage guarantees, (2) LLM-guided subgroup discovery, (3) black-box error rate evaluation, (4) finite-sample certification, (5) threshold calibration with compliance reporting.

Data splitting: We split the dataset into train (60%), validation (10%), test (15%), and audit (15%) sets. The audit set is held from all model development. The training and validation sets are used for ML classifiers. The

LLM uses few-shot examples from training set, and subgroup discovery uses only aggregate training statistics. This separation ensures the audit set remains unconstrained by any model development decisions.

Classifiers: We investigate different types of ML classifiers (LR, XGBoost, Random Forest, Equalized Odds reduction by FairLearn [1], and threshold optimizer) for both binary and multiclass classification settings. We also investigate seven open-source LLMs (3B to 30B parameters) served locally via Ollama (Llama 3.2, Qwen3, Gemma 3, GPT-OSS, Mistral 24B, Devstral 24B, and Nemotron), and image classifiers (SimpleCNN and ResNet18).

3.2 LLM-Guided Subgroup Discovery

Existing evaluation frameworks uses only predefined groups. As a result they miss proxy-based subgroups that may show disparities. To address this, we use a LLM-based *discover-then-confirm* mechanism. In the discovery phase, the LLM receives the dataset schema and statistics (feature names, types, data ranges, class balance, etc.) and returns subgroups. We use a deterministic prompt ¹ requesting a fixed number of subgroups, each with a reasoning field. In the next phase (confirmation), each candidate subgroup is certified using the Clopper-Pearson and Bonferroni mechanism described in Section 3.4, and we report smaller subgroups as *exploratory findings*. The framework treats discovery stage as hypothesis *generation* and certification as hypothesis *testing* on disjoint data to avoid selection bias that an unrestricted sweep of LLM-proposed subgroups would introduce.

3.3 Black-Box Error Rate Evaluation

After discovering the subgroups from the previous phase and training classifiers, the auditor module takes the classifier’s predictions on the held-out audit set and calculates the subgroup error-rate (TPR, FPR), and accuracy. Then we calculate the empirical gap statistics:

$$\Delta_{\text{FPR}}^{\text{emp}} = \max_{g, g'} |\text{FPR}_g - \text{FPR}_{g'}| \quad (1)$$

We also calculate $\Delta_{\text{TPR}}^{\text{emp}}$ in the same way. We use both $\Delta_{\text{FPR}}^{\text{emp}}$ and $\Delta_{\text{TPR}}^{\text{emp}}$ as assessment-report output. If a classifier’s positive prediction rate drops below 5% or exceeds 95%, the auditor raises a degeneracy flag and we do not certify these models as fair. This is because such classifiers can achieve near-zero gaps without meaningful predictive behavior. Also, the auditor only reads predictions on the audit set, so the anti-leakage guarantee from the data split still holds.

3.4 Finite-Sample Certificates

The main contribution of this framework is a finite-sample guarantee on the error-rate gap. We want to certify

$$\Pr \left(\max_{\substack{g, g' \in \mathcal{G} \\ g \neq g'}} |\text{FPR}_g - \text{FPR}_{g'}| \leq \epsilon \right) \geq 1 - \delta \quad (2)$$

where, δ is failure probability, and ϵ denotes tolerance rate.

¹Prompt can be found in the source code

3.4.1 Bonferroni Certificate (Clopper-Pearson)

Per-group confidence intervals: For group g with n_g^- actual negatives ($Y = 0$), we compute Clopper-Pearson exact intervals [8]:

$$L_g = B(\alpha'/2; \text{FP}_g, n_g^- - \text{FP}_g + 1) \quad (3)$$

$$U_g = B(1 - \alpha'/2; \text{FP}_g + 1, n_g^- - \text{FP}_g) \quad (4)$$

where $B(\cdot; a, b)$ is the Beta quantile function and $\alpha' = \delta/K$ is the Bonferroni-corrected significance level, L_g, U_g are lower and upper endpoints of the interval for group g . K is the number of groups under test, n_g^- the number of true negatives in group g and FP_g its false positive count.

Undefined rates: If a group has zero actual negatives ($n_g^- = 0$), FPR is undefined, such groups are excluded from FPR gap certification (same goes for TPR). If exclusion leaves fewer than two groups, certification is declared infeasible.

Certified bound: The worst-case gap bound is:

$$\Delta_{\text{FPR}} = \max_{g \neq g'} \max(U_g - L_{g'}, U_{g'} - L_g)$$

The certificate passes if $\Delta_{\text{FPR}} \leq \varepsilon$.

PROPOSITION 3.1 (BONFERRONI CERTIFICATE VALIDITY). *Let $\mathcal{G}$ be a set of K groups, and for each $g \in \mathcal{G}$ let $[L_g, U_g]$ be the Clopper-Pearson interval for FPR_g at significance level $\alpha' = \delta/K$. Define the certified bound*

$$\Delta_{\text{FPR}} = \max_{\substack{g, g' \in \mathcal{G} \\ g \neq g'}} \max(U_g - L_{g'}, U_{g'} - L_g).$$

Then

$$\Pr \left(\max_{\substack{g, g' \in \mathcal{G} \\ g \neq g'}} |\text{FPR}_g - \text{FPR}_{g'}| \leq \Delta_{\text{FPR}} \right) \geq 1 - \delta.$$

In particular, if $\Delta_{\text{FPR}} \leq \varepsilon$, then the classifier is certified at tolerance ε with confidence $1 - \delta$.

Proof sketch: Each Clopper-Pearson interval has coverage $\geq 1 - \alpha'$ where $\alpha' = \delta/K$. By Bonferroni's inequality over K groups, all intervals simultaneously cover with probability $\geq 1 - \delta$. When all cover, Δ_{FPR} upper-bounds the true gap. The full proof is given in the supplementary material.

3.4.2 Permutation-Based Hypothesis Test

The Bonferroni correction is conservative, especially for large K , as it treats each group-wise confidence interval as independent. We further introduce a permutation-based hypothesis test derived from classical permutation inference [12, 19].

Null hypothesis: We test the null hypothesis H_0 that the classifier has the same false positive rate across all groups. Under H_0 , group labels are exchangeable within the $Y = 0$ (actual negatives). We calculate the null distribution of Δ by permuting group labels among the true negatives only, with the predictions held fixed. The full procedure is given in Algorithm 1.

Algorithm 1 Permutation-Based FPR Gap Test**Require:** Predictions $\tilde{y}$, labels y , groups s , permutations B , tolerance ε , level δ

- 1: Compute observed gap $\Delta^{\text{obs}} = \max_{g \neq g'} |\text{FPR}_g - \text{FPR}_{g'}|$
- 2: Let $\mathcal{N} = \{i : y_i = 0\}$ {actual negatives only}
- 3: **for** $b = 1$ to B **do**
- 4: $s_{\mathcal{N}}^{(b)} \leftarrow \text{RandomPermutation}(s_{\mathcal{N}})$ {permute within $Y=0$ stratum}
- 5: Compute $\Delta^{(b)}$ from per-group FPR using $s_{\mathcal{N}}^{(b)}$
- 6: **end for**
- 7: $c_\delta \leftarrow Q_{1-\delta}(\{\Delta^{(1)}, \dots, \Delta^{(B)}\})$ {null critical value}
- 8: $p \leftarrow \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\Delta^{(b)} \geq \Delta^{\text{obs}}]$
- 9: **Pass** if $p \geq \delta$ (fail to reject H_0) **and** $c_\delta \leq \varepsilon$

Type I error control: Under H_0 , exchangeability within $Y = 0$ ensures that the permutation test controls the probability of falsely rejecting a fair classifier at the level δ [19]. Here the guarantee is conditional on H_0 (unlike Bonferroni’s unconditional CIs), so the permutation bound characterizes what the gap would be under equal FPRs, not an unconditional upper bound on the true gap. And because the implementation uses Monte Carlo permutations ($B = 10,000$) rather than full enumeration, the empirical p-value is an approximation. The bound is narrower because the permutation test captures the joint distribution of per-group FPRs, avoiding the union bound penalty. We recommend Bonferroni for formal certification and the permutation test for exploratory analysis.

3.4.3 Bootstrap Diagnostic

We also include a bootstrap diagnostic for exploratory purpose. Each of B , bootstrap draws resamples examples with replacement within each group and recomputes the worst-pair FPR gap, yielding bootstrap statistics $\Delta_{\text{FPR}}^{*(1)}, \dots, \Delta_{\text{FPR}}^{*(B)}$. From these draws we form two upper bounds at level $1-\delta$, the percentile bound $U_{\text{pct}} = Q_{1-\delta}(\Delta_{\text{FPR}}^*)$ and the basic bound $U_{\text{bas}} = 2\Delta^{\text{obs}} - Q_\delta(\Delta_{\text{FPR}}^*)$, where $Q_p(\cdot)$ denotes the p -quantile of the bootstrap draws. We report the more conservative of the two, $U = \max(U_{\text{pct}}, U_{\text{bas}})$, floored at zero.

Asymptotic coverage and limitations: Under standard regularity conditions, the bootstrap distribution of Δ_{FPR}^* converges to the sampling distribution of Δ_{FPR} . However, the max-gap functional $\Delta = \max_{g, g'} |\text{FPR}_g - \text{FPR}_{g'}|$ is directionally differentiable, so the standard bootstrap can be inconsistent at boundary configurations where two groups have nearly equal FPRs [10, 11]. The basic bootstrap interval corrects bias relative to the percentile method but does not achieve second-order accuracy for such non-smooth statistics [22]. For small groups ($n_g < 50$), the bootstrap may undercover, we recommend $n_g \geq 100$. We therefore present bootstrap values as an exploratory diagnostic, not a formal certificate.

3.5 Threshold Calibration and Compliance

When a classifier fails certification, we search per-group thresholds t_g to equalize FPR and report the TPR cost incurred for each group. The final stage then produces a structured report. The report is generated as both JSON (machine-readable) and plain text (human-readable), and contains seven sections: (1) *System Description*, (2) *Risk Classification*, (3) *Data Governance*, (4) *Fairness Assessment* with per-classifier FPR/TPR findings, (5) *Certification*

Status listing pass/fail outcomes per ε , (6) *Bias Mitigation* measures applied, and (7) *Recommendations* for compliance. A sample report is included in the released code.

3.6 Multi-Class Extension

Besides binary classification certificates, we extend our framework to multi-class classification as many real-world decision systems operate over multiple outcome categories. For each class $k \in \{0, \dots, K-1\}$, we binarize labels as $y^{(k)} = \mathbf{1}[y = k]$ and predictions as $\tilde{y}^{(k)} = \mathbf{1}[\tilde{y} = k]$, then compute FPR/TPR certificates on each binary sub-problem using the same Clopper-Pearson intervals.

Confidence adjustment: To maintain overall confidence $1 - \delta$ across K sub-problems, we apply an additional Bonferroni correction. Each class uses significance level $\alpha_k = \delta/K$, so within each class the per-group correction becomes $\delta/(K \cdot G)$ where G is the number of demographic groups. By union bound, all K class-level certificates hold simultaneously with probability $\geq 1 - \delta$.

Worst-case aggregation: The multi-class certificate reports the worst-case certified bounds across all K classes for both FPR and TPR:

$$\Delta_{\text{FPR}}^{\text{OvR}} = \max_{k=0}^{K-1} \Delta_{\text{FPR}}^{(k)}, \quad \Delta_{\text{TPR}}^{\text{OvR}} = \max_{k=0}^{K-1} \Delta_{\text{TPR}}^{(k)}.$$

The system passes if $\Delta_{\text{FPR}}^{\text{OvR}} \leq \varepsilon$ and $\Delta_{\text{TPR}}^{\text{OvR}} \leq \varepsilon$, meaning every class must satisfy both fairness tolerances.

4 Experiments

Datasets: We conduct experiments on eight benchmark datasets, *COMPAS*, *ACS Income*, *ACS Income 3-Class* (3 target classes: Low $\leq 30\text{K}$, Medium $30\text{K} - 75\text{K}$, High $> 75\text{K}$), *Credit default*, *LAW School*, *Bank Marketing*, *FairFace* (Image), *UTKFace* (Image). All datasets use a 60/10/15/15 stratified split.

Metrics: To evaluate our framework, we use the following metrics: FPR gap ($\max_{g,g'} |\text{FPR}_g - \text{FPR}_{g'}|$), TPR gap, certificate tightness, certification pass / fail at $\varepsilon \in \{0.05, 0.10, 0.15\}$ with 95% confidence, accuracy, and AUC. We run all ML experiments using 5 seeds (42, 123, 456, 789, 1024) for classifier training and report the mean and standard deviation. We keep the data split fixed for all the seeds to isolate model randomness from split randomness. As a result, deterministic classifiers (LR, XGBoost) demonstrate std of 0, whereas stochastic classifiers show non-zero variance due to their internal randomization.

Implementation details: Image classifiers (SimpleCNN, ResNet18) are trained for up to 500 epochs with early stopping on best validation accuracy (learning rate 10^{-3} , batch size 128). LLM experiments use Ollama with temperature 0.0 (deterministic) unless stated otherwise. The standard prompt template asks the model to “predict the binary outcome” given the individual’s profile and respond with JSON. Few-shot uses $k = 5$ total examples ($\lfloor k/2 \rfloor$ positive, $\lceil k/2 \rceil$ negative) drawn from the training set with a fixed seed. Variance experiments use 5 seeds with temperature 0.7. Permutation tests use $B = 10,000$ permutations and bootstrap uses $B = 10,000$ resamples.

4.1 Results

4.1.1 COMPAS: ML and LLM Classifier Fairness

The results are given in Table 2. We observe that unconstrained classifiers exhibit FPR gaps of 0.15 to 0.19 on COMPAS. XGBoost achieves the best accuracy of 0.668 but carries a 0.175 FPR gap. Fairlearn EO cuts the gap to

Table 2. ML classifier results on COMPAS (mean $\pm$ std, 5 seeds). Certified: pass/fail at $\varepsilon \in \{0.05, 0.10, 0.15\}$, 95% confidence. Fairlearn EO and ThresholdOpt both use logistic regression as the base estimator.

Classifier	Acc	FPR Gap	TPR Gap	Certified (FPR)		
				0.05	0.10	0.15
LR	0.645 $\pm$ 0.000	0.190 $\pm$ 0.000	0.259 $\pm$ 0.000	✗	✗	✗
XGBoost	0.668 $\pm$ 0.000	0.175 $\pm$ 0.000	0.203 $\pm$ 0.000	✗	✗	✗
RF	0.593 $\pm$ 0.001	0.150 $\pm$ 0.008	0.159 $\pm$ 0.007	✗	✗	✗
Fairlearn EO	0.546 $\pm$ 0.011	0.033 $\pm$ 0.028	0.044 $\pm$ 0.031	✗	✗	✗
ThresholdOpt	0.628 $\pm$ 0.008	0.041 $\pm$ 0.007	0.028 $\pm$ 0.021	✗	✗	✓

0.033, though at a cost of approximately 10 accuracy points. No classifier passes certification at $\varepsilon = 0.05$ with only 792 audit samples.

Table 8 reports LLM results on COMPAS across six models (3B to 30B parameters) at temperature 0.0. We observe that scaling improves accuracy but not fairness, FPR gaps range from 0.10 to 0.21 across 8B to 30B models with no consistent trend. Llama 3.2 (3B) achieves near-zero gaps but only because it predicts almost everyone as non-recidivist (TPR < 1%). Only this trivial 3B model passes certification. To further assess LLM stability, we run Mistral 24B across 5 seeds at temperature 0.7 on the full audit set (792 samples). Zero-shot FPR gaps are stable (0.250 $\pm$ 0.013, 92% of predictions unanimous across seeds). Few-shot shows higher variance (0.178 $\pm$ 0.032, only 39% unanimous), driven by both LLM sampling noise and few-shot example selection.

4.1.2 ACS Income: Large-Sample Certification

With 29,350 audit samples (37x COMPAS), all classifiers pass certification at all ε levels with certified bounds 0.020 to 0.029 (full ML and LLM tables in the supplementary material). FPR gaps are smaller on the LLM task relative to COMPAS, so fairness gaps appear to be task dependent rather than a fixed property of the model.

4.1.3 Additional Tabular Datasets

Table 3 summarizes three datasets. On *Credit Default*, intersectional certification ($K = 4$) fails while single-attribute passes. On *Law School*, FPR gaps reach 0.375, which is the most severe disparity we observe across all datasets. On *Bank Marketing*, gaps are negligible.

4.1.4 Image Domain Certification

We also apply the auditor to two vision classifiers (Table 4). On FairFace, SimpleCNN passes certification in 4 of 5 seeds, the one failing seed points to training seed variance, not a systematic bias. On *UTKFace*, FPR gaps are small (0.022) but TPR gaps are large (0.157). This shows the importance of measuring both error rate types separately.

4.1.5 Multi-Class Certification

We list the multiclass certification results on ACS Income with 3 income classes (Low, Medium, High) in Table 5. For each class k , we binarize (k vs. rest) and run the standard Clopper-Pearson certificate with Bonferroni correction across $K = 3$ classes. XGBoost passes certification at $\varepsilon = 0.05$ for all classes, with the largest certified bound on

Table 3. Additional tabular results (mean $\pm$ std, 5 seeds).

Dataset	Classifier	Acc	FPR Gap	Cert@0.05	Audit n
Credit Def.	XGBoost	0.818	0.018 $\pm$ 0.000	✓	4,500
	ThresholdOpt	0.802	0.011 $\pm$ 0.001	✓	
Law School	XGBoost	0.900	0.375 $\pm$ 0.000	✗	3,120
	ThresholdOpt	0.890	0.000 $\pm$ 0.000	✓	
Bank Mktg	XGBoost	0.896	0.002 $\pm$ 0.000	✓	6,782
	ThresholdOpt	0.883	0.001 $\pm$ 0.000	✓	

Table 4. Image classifier results (mean $\pm$ std, 5 seeds).

Dataset	Model	Acc	AUC	FPR Gap	TPR Gap	Cert@0.05
FairFace	SimpleCNN	0.828 $\pm$ 0.004	0.913	0.010 $\pm$ 0.010	0.014 $\pm$ 0.007	4/5
	ResNet18	0.880 $\pm$ 0.002	0.954	0.017 $\pm$ 0.008	0.012 $\pm$ 0.008	3/5
UTKFace	SimpleCNN	0.878 $\pm$ 0.001	0.945	0.029 $\pm$ 0.007	0.154 $\pm$ 0.010	1/5
	ResNet18	0.892 $\pm$ 0.003	0.954	0.022 $\pm$ 0.008	0.157 $\pm$ 0.011	2/5

Table 5. Multi-class certification on ACS Income 3-Class (mean $\pm$ std, 5 seeds). Per-class FPR gap bounds via Clopper-Pearson with Bonferroni across $K=3$ classes and $G=2$ groups (sex). Worst: class with largest bound.

Classifier	Acc	Per-Class FPR Bound			Worst	Cert@0.05
		Low	Med	High		
LR	0.626 $\pm$ 0.000	0.023 $\pm$ 0.000	0.053 $\pm$ 0.000	0.036 $\pm$ 0.000	Med	✗
XGBoost	0.654 $\pm$ 0.000	0.023 $\pm$ 0.000	0.042 $\pm$ 0.000	0.022 $\pm$ 0.000	Med	✓
RF	0.621 $\pm$ 0.000	0.030 $\pm$ 0.001	0.037 $\pm$ 0.000	0.024 $\pm$ 0.001	Med	✓

the Medium class 0.042. However, LR and RF fail at $\epsilon = 0.05$ due to the Medium class but pass at $\epsilon = 0.10$. This shows that the medium income range (30K to 75K) is the worst-certified class for all three classifiers.

Additionally, Table 6 compares all three certification methods per class (XGBoost, seed 42). The results are consistent with our binary findings. The Medium class is the hardest to certify across all methods, reflecting income-range disparities between demographic groups.

4.1.6 Subgroup Discovery and Calibration

LLM-discovered subgroups on COMPAS (full table in the supplementary material) reveal “Caucasian females with high priors” (FPR=1.0, $n = 13$) and “high decile scores” (FPR=0.735, $n = 147$), both exceeding the overall Black FPR 0.342. Note that the $n = 13$ subgroup is too small for formal certification ($n_g < 30$) and should be treated as an exploratory finding, whereas the $n = 147$ subgroup meets the minimum for certification. Threshold

Table 6. Three-method comparison per OvR class (XGBoost, ACS Income 3-Class, $\epsilon=0.05$). See Table 7 caption for column definitions.

Class (OvR)	Bonf.	Perm. CV	Boot. [†]
Low ($\leq 30K$)	0.023	0.013	0.014
Medium (30K–75K)	0.042	0.015	0.032
High ($> 75K$)	0.022	0.011	0.015

Table 7. Three-method comparison (FPR gap, $\epsilon=0.05$, XGBoost). Bonf.: unconditional upper confidence bound on true gap. Perm. CV: critical value of the null distribution under H_0 (a test statistic, not a confidence bound). Boot.[†]: exploratory asymptotic diagnostic (not a formal bound; see Section 3). Lower values indicate narrower estimates.

Dataset	Groups	K	Bonf.	Perm. CV	Boot. [†]
COMPAS	Primary	2	0.305	0.085	0.242
	Intersect.	4	0.439	0.196	0.344
ACS Inc.	Primary	2	0.028	0.011	0.019
	Intersect.	4	0.048	0.021	0.031
Cr. Def.	Primary	2	0.043	0.015	0.031
	Intersect.	4	0.064	0.027	0.044
Law Sch.	Primary	2	0.518	0.098	0.456
Bank	Primary	2	0.012	0.006	0.007
	Intersect.	4	0.027	0.013	0.018

calibration (supplementary) shows that equalizing FPR costs 9.8% TPR for Black defendants, quantifying the FPR-TPR trade-off [14].

4.1.7 Method Comparison

Table 7 compares all three methods across datasets and group configurations, and Figure 2 visualises the same values. On *Credit Default intersectional* ($K = 4$, groups pre-specified as sex x age binarized at median, not LLM-discovered), Bonferroni 0.064 fails at $\epsilon = 0.05$, while the permutation value 0.027 and bootstrap diagnostic 0.044 show narrower values. The permutation value reflects the null distribution width, not an unconditional CI. The bootstrap diagnostic is unconditional but asymptotic and lacks formal consistency guarantees for the non-smooth max-gap functional. For formal regulatory certification, Bonferroni remains the only defensible choice, the other methods are useful for identifying cases where Bonferroni conservatism drives certification failure.

4.1.8 Model Scale Ablation (3B to 30B)

Table 8 and Figure 3 show a non-monotonic relationship between scale and fairness across six open-weight models. Llama 3.2 (3B) has near-zero FPR gaps but random-level accuracy, 8B to 30B models show gaps of 0.048 to 0.24 (few-shot) with no consistent scale trend. Accuracy, in contrast, improves significantly with scale. Only the 3B model passes certification, trivially due to near-random predictions.

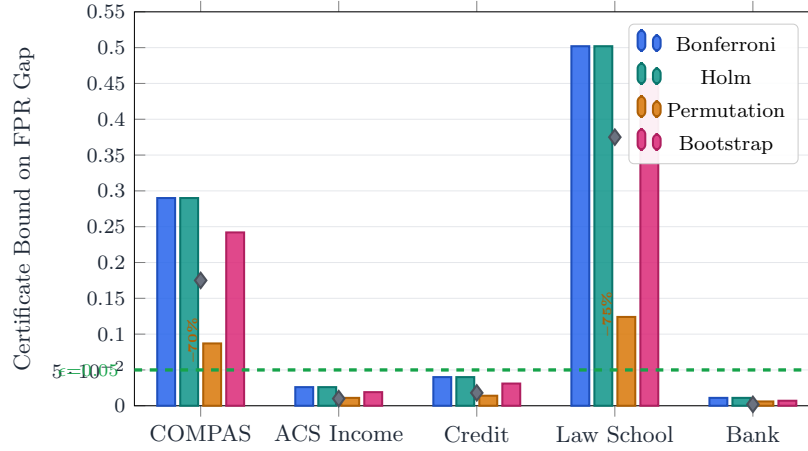


Fig. 2. Values across five datasets (primary groups, XGBoost). Permutation test (amber) yields narrower critical values than Bonferroni bounds (blue) but with guarantees conditional on H_0 . Bootstrap (rose) is an exploratory diagnostic. Diamonds mark empirical gaps. The green line marks $\epsilon=0.05$.

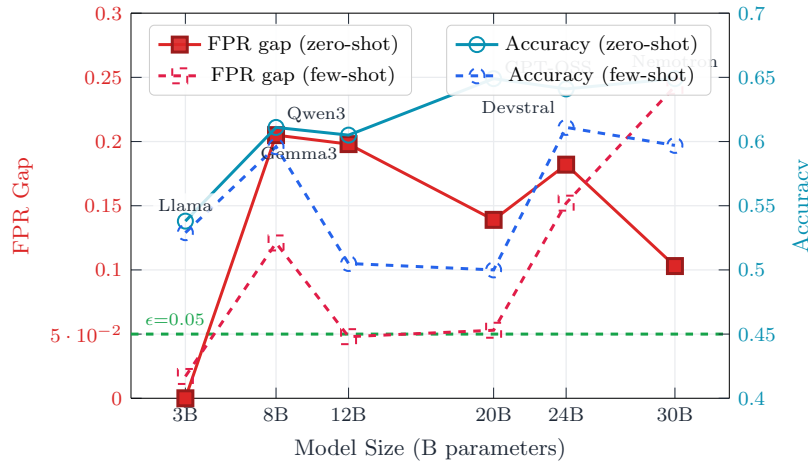


Fig. 3. Model scale vs. fairness on COMPAS (6 models, 3B–30B). Left axis (red): FPR gap; right axis (cyan): accuracy. Larger models improve accuracy but not fairness. The green line marks $\epsilon = 0.05$.

4.1.9 LLM Robustness: Prompt and Temperature Ablation

Prompt sensitivity and certification implications: Three prompt templates on Mistral 24B at temperature 0.1 show FPR gaps of 0.097 to 0.266 (Table 9). For reference, Mistral at the default temperature 0.0 with the standard template produces an FPR gap of 0.259. Clopper-Pearson certification outcomes depend on the prompt: at $\epsilon = 0.25$, the neutral template passes (bound=0.235) while standard and detailed fail. The robustness margin, defined as

Table 8. Model scale ablation on COMPAS: 6 open-weight models from 3B to 30B parameters, all served locally via Ollama. FPR gap does not decrease monotonically with scale; the smallest model (3B) has near-zero gap but random-level accuracy.

Model	Size	Mode	Acc	FPR Gap	TPR Gap	Bonf.	Perm. CV	Boot. [†]
Llama 3.2	3B	zero	0.538	0.000	0.005	✓	✓	✓
		few	0.529	0.017	0.013	✗	✓	✓
Qwen3	8B	zero	0.611	0.205	0.125	✗	✗	✗
		few	0.596	0.121	0.080	✗	✗	✗
Gemma 3	12B	zero	0.605	0.198	0.137	✗	✗	✗
		few	0.505	0.048	0.006	✗	✗	✗
GPT-OSS	20B	zero	0.649	0.139	0.216	✗	✗	✗
		few	0.500	0.053	0.081	✗	✗	✗
Devstral	24B	zero	0.641	0.181	0.173	✗	✗	✗
		few	0.611	0.152	0.096	✗	✗	✗
Nemotron	30B	zero	0.649	0.103	0.211	✗	✗	✗
		few	0.597	0.243	0.110	✗	✗	✗

Table 9. Prompt sensitivity and certification: Mistral 24B on COMPAS (zero-shot, temperature 0.1).

Template	Acc	FPR Gap	Cert. Bound	Cert@0.25
Standard	0.641	0.266	0.394	✗
Neutral	0.591	0.097	0.235	✓
Detailed	0.648	0.238	0.366	✗

the gap between the best and worst-case certified bounds, is 0.159. A temperature ablation in the supplementary material shows that FPR gaps are stable across temperatures 0.0 to 0.5 (0.259 to 0.270).

5 Discussion

We recommend Bonferroni for formal regulatory certification due to its unconditional, finite-sample guarantees. The permutation test is appropriate for exploratory testing but its narrower value reflects H_0 -conditional reasoning, not an unconditional certificate. The bootstrap provides an unconditional but asymptotic bound.

LLM fairness requires per-domain evaluation: Mistral’s FPR gap drops from 0.259 (COMPAS) to 0.009 (ACS Income, Fig. 4). General LLM fairness benchmarks are insufficient for regulatory compliance, and each deployment domain requires its own assessment.

Exchangeability conditions: The permutation test assumes group-label exchangeability among actual negatives under H_0 , conditioning on true negatives handles differing base rates. It can be anti-conservative under non-common populations or unobserved confounders correlated with group labels conditional on Y . Bonferroni remains valid without this assumption.

Supporting EU AI Act compliance: The Act classifies credit scoring (Annex III, Area 5(b)) and criminal-justice AI (Area 8(a)) as high-risk, requiring conformity assessments (Art. 43) [25]. **FairCert is not itself a**

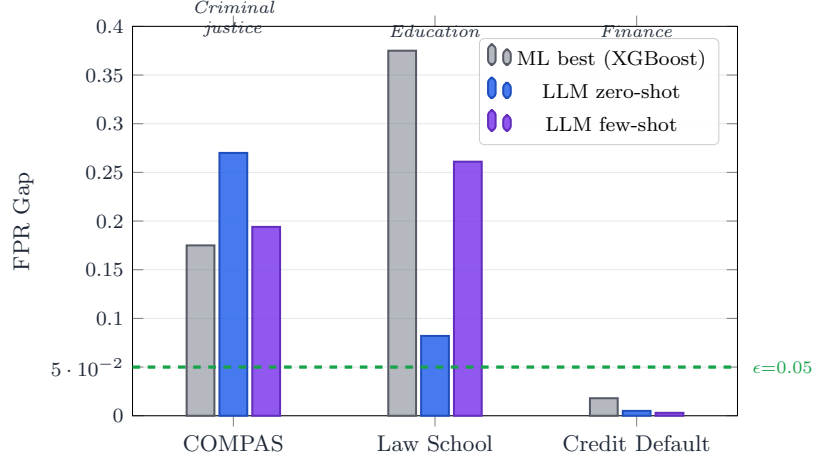


Fig. 4. Task-dependent LLM fairness across three domains (Mistral 24B). Criminal justice shows large gaps. Credit Default (finance) is near-certifiable.

conformity assessment but supplies auditing evidence supporting three obligations: **Art. 9 (risk management)** via FPR/TPR certificates that quantify the risk of unequal treatment with formal confidence guarantees, **Art. 10 (data governance)** via the sample complexity that specifies minimum per-group negatives (FPR) or positives (TPR), with prevalence-adjusted conversion to total audit size, and **Art. 15 (accuracy)** by accepting any ϵ set by notified bodies or domain regulators.

Limitations: The permutation test is $O(B \cdot n)$ (≈ 45 s on ACS Income). LLM inference is slow (≈ 2 samples/s), so variance study (5 seeds, $T = 0.7$) covers only Mistral 24B. LLM-discovered subgroups require expert review. We flag positive-prediction rates outside $[5\%, 95\%]$, with joint minimum-utility conditions left to future work.

6 Conclusion

FairCert provides finite-sample fairness guarantees for binary and multi-class classifiers. Bonferroni supplies unconditional guarantees for formal compliance, permutation and bootstrap serve as exploratory tools. We also handle the multi-class setting with controlled family-wise error. Evaluation on eight datasets demonstrates that the certificates are computable, informative, applicable across domains, and produce evaluation evidence relevant to EU AI Act obligations. In future work, we will look into calibration certificates and bootstrap coverage for the max-gap functional.

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